MnF$_2$: From the Microscopic Hamiltonian to LSWT, Bosonic BdG, and the Two-Mode Projection

A unified derivation of J_1–J_7, single-ion anisotropy, long-range DDI, magnetic field, and the projected two-mode model

Using the full-position Fourier gauge and a two-sublattice Néel convention, the calculation proceeds as

$$\mathcal{H}_{\text{micro}} \to (A_{\mathbf{k}}, B_{\mathbf{k}}) \to \mathcal{H}_{\text{BdG}}(\mathbf{k}) \to \omega_n(\mathbf{k}) \to H^{(2)}_{\text{eff}}(\mathbf{Q}, B) \to B^*(\mathbf{Q}),\ C(B).$$

Here $\lambda_{\text{AM}}(\mathbf{k})$ is the microscopic exchange form factor, $\Lambda_{\mathbf{Q}}$ is the signed diagonal detuning in the target positive-frequency doublet, and $D_{\mathbf{Q}}$ is the gauge-invariant transverse mixing obtained by projecting the full DDI operator.

1. Lattice, magnetic order, and local frames

Take the tetragonal lattice vectors

$$\mathbf{a}_1 = (a, 0, 0), \qquad \mathbf{a}_2 = (0, b, 0), \qquad \mathbf{a}_3 = (0, 0, c), \tag{1}$$

and the two Mn sublattice positions

$$\mathbf{r}_A = (0, 0, 0), \qquad \mathbf{r}_B = \left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right). \tag{2}$$

The ordered state is $A \uparrow, B \downarrow$ with the Néel vector along the crystallographic c axis. To use a common Holstein–Primakoff (HP) expansion on both sublattices, define

$$R_A = I_3, \qquad R_B = \text{diag}(-1, 1, -1), \tag{3}$$

with slope

$$\mathbf{S}_{i\mu} = R_\mu \widetilde{\mathbf{S}}_{i\mu}, \qquad \mu = A, B. \tag{4}$$

Both ordered moments then point along local $+\widetilde{z}$.

In the full-position Fourier gauge, the displacement of a bond from sublattice μ to ν is

$$\boldsymbol{\delta} = \mathbf{R} + \mathbf{r}_\nu - \mathbf{r}_\mu, \tag{5}$$

with Fourier phase $e^{i\mathbf{k}\cdot\boldsymbol{\delta}}$. Equivalently,

$$\widetilde{c}_{\mu,\mathbf{R}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} c_{\mu,\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{R}+\mathbf{r}_\mu)}. \tag{6}$$

This is the full-position gauge, with the basis positions included in the bond displacement $\boldsymbol{\delta}$.

2. Microscopic Hamiltonian

The complete model is

$$\boxed{\mathcal{H} = \mathcal{H}_{\text{ex}} + \mathcal{H}_{\text{SIA}} + \mathcal{H}_{\text{dip}} + \mathcal{H}_Z}. \tag{7}$$

The individual terms are

$$\mathcal{H}_{\text{ex}} = \sum_{(i\mu, j\nu)\in\mathcal{B}_{\text{ex}}} J_{i\mu,j\nu}\, \mathbf{S}_{i\mu} \cdot \mathbf{S}_{j\nu}, \tag{8}$$

$$\mathcal{H}_{\text{SIA}} = \sum_{i\mu} \mathbf{S}_{i\mu}^{\mathsf{T}} \mathsf{A}_\mu \mathbf{S}_{i\mu}, \tag{9}$$

$$\mathcal{H}_{\text{dip}} = C_{\text{dip}} \sum_{i\mu<j\nu} \left[\frac{\mathbf{S}_{i\mu} \cdot \mathbf{S}_{j\nu}}{r^3} - 3\frac{(\mathbf{S}_{i\mu} \cdot \mathbf{r})(\mathbf{S}_{j\nu} \cdot \mathbf{r})}{r^5} \right], \tag{10}$$

$$\mathcal{H}_Z = -g\mu_{\text{B}} B \sum_{i\mu} S^z_{i\mu}, \qquad B \parallel c, \tag{11}$$

where $\mathbf{r} = \mathbf{r}_{j\nu} - \mathbf{r}_{i\mu}$ and

$$C_{\text{dip}} = \frac{\mu_0}{4\pi} (g\mu_{\text{B}})^2. \tag{12}$$

The unit conversion is $\mu_0\mu_B^2/(4\pi\,\text{Å}^3) = 0.0536815\,\text{meV}$.

3. HP expansion and generic BdG rules

In each sublattice local frame,

$$\widetilde{S}^z = S - c^\dagger c, \qquad \widetilde{S}^+ = \sqrt{2S}\,c, \qquad \widetilde{S}^- = \sqrt{2S}\,c^\dagger, \tag{13}$$

or

$$\widetilde{S}^x = \sqrt{\frac{S}{2}}(c + c^\dagger), \qquad \widetilde{S}^y = -i\sqrt{\frac{S}{2}}(c - c^\dagger). \tag{14}$$

Terms are retained through quadratic order, $O(S)$ in LSWT.

3.1 Generic bilinear tensor interaction

For a global tensor coupling

$$\mathbf{S}_{i\mu}^{\mathsf{T}} \mathsf{K}_{\mu\nu}^{\mathrm{glob}} \mathbf{S}_{j\nu}, \tag{15}$$

first rotate it to the local frames,

$$\mathsf{K}_{\mu\nu}^{\mathrm{loc}} = R_{\mu}^{\mathsf{T}} \mathsf{K}_{\mu\nu}^{\mathrm{glob}} R_{\nu}. \tag{16}$$

Writing the components as $K_{\alpha\beta}$, the transverse HP coefficients are

$$t_{\mu\nu} = \frac{S}{2}\left[(K_{xx} + K_{yy}) + i(K_{yx} - K_{xy})\right], \tag{17}$$

$$p_{\mu\nu} = \frac{S}{2}\left[(K_{xx} - K_{yy}) - i(K_{xy} + K_{yx})\right]. \tag{18}$$

The longitudinal contribution is set by K_{zz}; t enters the normal block A and p the anomalous/pairing block B.

3.2 Same-sublattice isotropic exchange

For a same-sublattice isotropic bond,

$$J\widetilde{\mathbf{S}}_{\mathbf{R}} \cdot \widetilde{\mathbf{S}}_{\mathbf{R}+\boldsymbol{\delta}}, \tag{19}$$

the quadratic part is

$$JS^2 + JS\left(c_{\mathbf{R}}^{\dagger} c_{\mathbf{R}+\boldsymbol{\delta}} + c_{\mathbf{R}+\boldsymbol{\delta}}^{\dagger} c_{\mathbf{R}} - n_{\mathbf{R}} - n_{\mathbf{R}+\boldsymbol{\delta}}\right). \tag{20}$$

After Fourier transformation,

$$\mathcal{H}_{J,\boldsymbol{\delta}}^{(2)} = 2JS \sum_{\mathbf{k}} [\cos(\mathbf{k}\cdot\boldsymbol{\delta}) - 1] c_{\mathbf{k}}^{\dagger} c_{\mathbf{k}}. \tag{21}$$

Same-sublattice isotropic exchange contributes only to the normal block and generates no pairing.

3.3 Opposite-sublattice isotropic exchange

For an A–B bond,

$$R_A^{\mathsf{T}} I R_B = \mathrm{diag}(-1, 1, -1), \tag{22}$$

so

$$\mathbf{S}_A \cdot \mathbf{S}_B = -\widetilde{S}_A^x \widetilde{S}_B^x + \widetilde{S}_A^y \widetilde{S}_B^y - \widetilde{S}_A^z \widetilde{S}_B^z. \tag{23}$$

At quadratic order,

$$J\mathbf{S}_A \cdot \mathbf{S}_B = -JS^2 + JS(n_A + n_B) - JS(ab + a^{\dagger} b^{\dagger}). \tag{24}$$

An antiferromagnetic inter-sublattice bond gives the same $+JS$ restoring contribution to both diagonal normal elements and generates anomalous pairing.

4. Exchange paths J_1–J_7: term-by-term derivation

Define

$$x = k_x a, \qquad y = k_y b, \qquad z = k_z c. \tag{25}$$

The independent bonds used in the two-sublattice calculation are listed below. Same-sublattice displacements are shown only once, with $-\boldsymbol{\delta}$ supplied by the Hermitian conjugate; all $\pm$ combinations are listed explicitly for the inter-sublattice J_2 and J_5 bonds.

Exchange	Type	Independent displacement $\boldsymbol{\delta}$	Coordination
J_1	same sublattice	$(0, 0, c)$	2
J_2	A–B	$(\pm a/2, \pm b/2, \pm c/2)$	8
J_3	same sublattice	$(a, 0, 0), (0, b, 0)$	4
J_4	same sublattice	$(a, 0, \pm c), (0, b, \pm c)$	8
J_5	A–B	$(\pm a/2, \pm b/2, \pm 3c/2)$	8
J_6	same sublattice	$(0, 0, 2c)$	2
J_{7a}, J_{7b}	same sublattice	$(a, -b, 0), (a, b, 0)$ with labels interchanged on B	$2 + 2$

4.1 J_1: same-sublattice exchange along c

$$A^{(1)}(\mathbf{k}) = 2SJ_1(\cos z - 1). \tag{26}$$

The contribution is identical on A and B.

4.2 J_2: nearest inter-sublattice body-diagonal exchange

The structure factor is

$$\gamma_2(\mathbf{k}) = \sum_{s_x,s_y,s_z=\pm1} e^{\frac{i}{2}(s_x x+s_y y+s_z z)} \tag{27}$$

$$= 8\cos\frac{x}{2}\cos\frac{y}{2}\cos\frac{z}{2}. \tag{28}$$

Each inter-sublattice bond contributes a diagonal restoring field $+SJ_2$, giving

$$A^{(2)}_{AA} = A^{(2)}_{BB} = 8SJ_2, \tag{29}$$

while the pairing term is

$$B^{(2)}_{AB} = B^{(2)}_{BA} = -8SJ_2\cos\frac{x}{2}\cos\frac{y}{2}\cos\frac{z}{2}. \tag{30}$$

4.3 J_3: in-plane axial same-sublattice exchange

$$A^{(3)}(\mathbf{k}) = 2SJ_3(\cos x + \cos y - 2). \tag{31}$$

4.4 J_4: mixed in-plane and c-axis same-sublattice exchange

$$A^{(4)}(\mathbf{k}) = 2SJ_4\big[\cos(x+z) + \cos(x-z) + \cos(y+z) + \cos(y-z) - 4\big] \tag{32}$$

$$= 4SJ_4\left[(\cos x + \cos y)\cos z - 2\right]. \tag{33}$$

4.5 J_5: longer-range inter-sublattice exchange

$$\gamma_5(\mathbf{k}) = \sum_{s_x,s_y,s_z=\pm1} e^{i(s_x x/2+s_y y/2+3s_z z/2)} \tag{34}$$

$$= 8\cos\frac{x}{2}\cos\frac{y}{2}\cos\frac{3z}{2}, \tag{35}$$

which gives

$$A^{(5)}_{AA} = A^{(5)}_{BB} = 8SJ_5, \tag{36}$$

$$B^{(5)}_{AB} = B^{(5)}_{BA} = -8SJ_5\cos\frac{x}{2}\cos\frac{y}{2}\cos\frac{3z}{2}. \tag{37}$$

4.6 J_6: second same-sublattice distance along c

$$A^{(6)}(\mathbf{k}) = 2SJ_6(\cos 2z - 1). \tag{38}$$

4.7 J_{7a}, J_{7b}: altermagnetic diagonal detuning

Define

$$\boldsymbol{\delta}_a = (a, -b, 0), \qquad \boldsymbol{\delta}_b = (a, b, 0), \qquad \delta J_7 = J_{7b} - J_{7a}, \tag{39}$$

with the J_{7a} and J_{7b} bond labels interchanged on sublattice B. Then

$$A^{(7)}_A(\mathbf{k}) = 2S\left\{J_{7a}[\cos(x-y) - 1] + J_{7b}[\cos(x+y) - 1]\right\}, \tag{40}$$

$$A^{(7)}_B(\mathbf{k}) = 2S\left\{J_{7b}[\cos(x-y) - 1] + J_{7a}[\cos(x+y) - 1]\right\}. \tag{41}$$

With

$$\bar{J}_7 = \frac{J_{7a} + J_{7b}}{2}, \tag{42}$$

we obtain

$$A^{(7)}_A = 4S\bar{J}_7(\cos x\cos y - 1) - 2S\delta J_7\sin x\sin y, \tag{43}$$

$$A^{(7)}_B = 4S\bar{J}_7(\cos x\cos y - 1) + 2S\delta J_7\sin x\sin y. \tag{44}$$

Subtracting the two expressions gives

$$A^{(7)}_A - A^{(7)}_B = -4S\delta J_7\sin x\sin y. \tag{45}$$

Define

$$\lambda_{\mathrm{AM}}(\mathbf{k}) = 4S\delta J_7\sin(k_x a)\sin(k_y b), \tag{46}$$

with slope

$$A_A - A_B = -\lambda_{\mathrm{AM}}(\mathbf{k}). \tag{47}$$

For $\mathbf{Q} = (h, k, l)$ in reciprocal-lattice units,

$$\lambda_{\mathrm{AM}}(h, k, l) = 4S\delta J_7\sin(2\pi h)\sin(2\pi k), \tag{48}$$

and

$$\lambda_{\mathrm{AM}}(C_{4z}\mathbf{Q}) = -\lambda_{\mathrm{AM}}(\mathbf{Q}). \tag{49}$$

5. Exchange BdG matrix from J_1–J_7

All sublattice-even normal terms combine into

$$A_0(\mathbf{k}) = S\Big\{2J_1(\cos z - 1) + 2J_3(\cos x + \cos y - 2) \tag{50}$$

$$+ 4J_4[(\cos x + \cos y)\cos z - 2] + 2J_6(\cos 2z - 1) \tag{51}$$

$$+ 4\bar{J}_7(\cos x \cos y - 1) + 8J_2 + 8J_5\Big\}. \tag{52}$$

The inter-sublattice pairing is

$$\Gamma(\mathbf{k}) = -8S\cos\frac{x}{2}\cos\frac{y}{2}\left(J_2\cos\frac{z}{2} + J_5\cos\frac{3z}{2}\right). \tag{53}$$

The exchange-only normal and pairing blocks are

$$A_{\mathrm{ex}}(\mathbf{k}) = \begin{pmatrix} A_0 - \lambda_{\mathrm{AM}}/2 & 0 \\ 0 & A_0 + \lambda_{\mathrm{AM}}/2 \end{pmatrix}, \qquad B_{\mathrm{ex}}(\mathbf{k}) = \begin{pmatrix} 0 & \Gamma \\ \Gamma & 0 \end{pmatrix}. \tag{54}$$

In the Nambu basis

$$\Psi_{\mathbf{k}} = (a_{\mathbf{k}}, b_{\mathbf{k}}, a^\dagger_{-\mathbf{k}}, b^\dagger_{-\mathbf{k}})^\top, \tag{55}$$

$$H^{\mathrm{BdG}}_{\mathrm{ex}}(\mathbf{k}) = \begin{pmatrix} A_{\mathrm{ex}}(\mathbf{k}) & B_{\mathrm{ex}}(\mathbf{k}) \\ B^\dagger_{\mathrm{ex}}(\mathbf{k}) & A^*_{\mathrm{ex}}(\mathbf{k}) \end{pmatrix}. \tag{56}$$

With DDI, SIA, and field switched off, the common positive-frequency center is

$$\omega_0(\mathbf{k}) = \sqrt{A_0^2(\mathbf{k}) - \Gamma^2(\mathbf{k})}, \tag{57}$$

and the two positive frequencies can be written as

$$\omega_\pm(\mathbf{k}) = \omega_0(\mathbf{k}) \pm \frac{\lambda_{\mathrm{AM}}(\mathbf{k})}{2}, \tag{58}$$

up to the overall branch-ordering convention. We take

$$\Lambda_{\mathbf{Q}} = -\lambda_{\mathrm{AM}}(\mathbf{Q}). \tag{59}$$

6. Single-ion anisotropy in LSWT

6.1 Axial scalar SIA

For

$$\mathcal{H}_{\mathrm{SIA}} = D_c \sum_i (S_i^z)^2, \tag{60}$$

we have locally

$$(\widetilde{S}^z)^2 = (S-n)^2 = S^2 - 2Sn + O(S^0), \tag{61}$$

and therefore

$$A^{\mathrm{axial}}_{\mathrm{SIA}} = -2SD_c I_2, \qquad B^{\mathrm{axial}}_{\mathrm{SIA}} = 0. \tag{62}$$

This corresponds to the code operation 'A += -2*S*Dc'; the production model uses $D_c = 0$.

6.2 General quadratic SIA tensor

More generally,

$$\mathcal{H}_{\mathrm{SIA}} = \sum_{i\mu} \mathbf{S}^\top_{i\mu} \mathsf{A}_\mu \mathbf{S}_{i\mu}, \qquad \widetilde{\mathsf{A}}_\mu = R^\top_\mu \mathsf{A}_\mu R_\mu. \tag{63}$$

At a stable ordered direction, the linear boson terms vanish. The quadratic contributions are

$$A_{\mathrm{SIA},\mu} = S(\widetilde{A}_{xx} + \widetilde{A}_{yy} - 2\widetilde{A}_{zz}), \tag{64}$$

$$B_{\mathrm{SIA},\mu} = S\left[(\widetilde{A}_{xx} - \widetilde{A}_{yy}) - 2i\widetilde{A}_{xy}\right], \tag{65}$$

The quadratic blocks are

$$A_{\mathrm{SIA}} = \mathrm{diag}(A_{\mathrm{SIA},A}, A_{\mathrm{SIA},B}), \qquad B_{\mathrm{SIA}} = \mathrm{diag}(B_{\mathrm{SIA},A}, B_{\mathrm{SIA},B}). \tag{66}$$

For the microscopic SIA robustness check, the two ideal-*mmm* site tensors in a common crystallographic frame are approximately

$$\mathsf{A}_A = \mathrm{diag}(-0.03, 0.53, -0.50)\,\mu\mathrm{eV}, \tag{67}$$

$$\mathsf{A}_B = \mathrm{diag}(0.53, -0.03, -0.50)\,\mu\mathrm{eV}. \tag{68}$$

The two tensors are related by the x/y interchange between the C_{4z}-related sublattices and do not generate the same sublattice-odd diagonal detuning as δJ_7. The numerical check gives a Γ gap of about 1.14345 meV for exchange+DDI and 1.16563 meV after adding this SIA; the main calculation uses the minimal exchange+DDI model.

7. Zeeman term and finite-field detuning

For $B \parallel c$,

$$\mathcal{H}_Z = -g\mu_B B\left(\sum_A S^z_A + \sum_B S^z_B\right). \tag{69}$$

Because $S^z_A = \widetilde{S}^z_A$ and $S^z_B = -\widetilde{S}^z_B$, the constants cancel and the quadratic term is

$$\mathcal{H}^{(2)}_Z = g\mu_B B\sum_{\mathbf{k}} a^\dagger_{\mathbf{k}} a_{\mathbf{k}} - g\mu_B B\sum_{\mathbf{k}} b^\dagger_{\mathbf{k}} b_{\mathbf{k}}. \tag{70}$$

The normal block acquires

$$A_Z(B) = g\mu_B B\,\sigma_z, \tag{71}$$

and the signed energy detuning between the two diabatic modes changes by

$$h(B) = 2g\mu_B B. \tag{72}$$

8. Long-range DDI: from the real-space tensor to BdG

8.1 Momentum-space dipolar tensor

Define

$$\mathsf{T}(\mathbf{r}) = C_{\text{dip}}\left(\frac{I_3}{r^3} - 3\frac{\mathbf{r}\mathbf{r}^\mathsf{T}}{r^5}\right), \tag{73}$$

and

$$\mathsf{D}_{\mu\nu}(\mathbf{Q}) = \sum_{\mathbf{R}}{}' \mathsf{T}(\mathbf{R}+\mathbf{r}_\nu-\mathbf{r}_\mu)e^{i\mathbf{Q}\cdot(\mathbf{R}+\mathbf{r}_\nu-\mathbf{r}_\mu)}. \tag{74}$$

The prime excludes the self pair $\mu = \nu, \mathbf{R} = 0$. Because the $1/r^3$ dipolar sum is conditionally convergent, the production calculation uses Ewald splitting.

8.2 Ewald decomposition

Let η be the Ewald parameter and $\mathbf{r} = \mathbf{R} + \mathbf{r}_\nu - \mathbf{r}_\mu$. The screened real-space tensor is

$$\mathsf{T}^{\text{real}}(\mathbf{r}) = C_{\text{dip}}\left[c_\delta(r)I_3 - c_{rr}(r)\,\mathbf{r}\mathbf{r}^\mathsf{T}\right], \tag{75}$$

with

$$c_\delta(r) = \frac{\operatorname{erfc}(\eta r)}{r^3} + \frac{2\eta}{\sqrt{\pi}}\frac{e^{-\eta^2 r^2}}{r^2}, \tag{76}$$

$$c_{rr}(r) = \frac{3\operatorname{erfc}(\eta r)}{r^5} + \frac{6\eta}{\sqrt{\pi}}\frac{e^{-\eta^2 r^2}}{r^4} + \frac{4\eta^3}{\sqrt{\pi}}\frac{e^{-\eta^2 r^2}}{r^2}. \tag{77}$$

The reciprocal-space part is

$$\mathsf{D}^{\text{recip}}_{\mu\nu}(\mathbf{Q}) = C_{\text{dip}}\frac{4\pi}{V}\sum_{\mathbf{G}}^{\star} e^{-K^2/(4\eta^2)}\frac{\mathbf{K}\mathbf{K}^\mathsf{T}}{K^2}e^{i\mathbf{K}\cdot(\mathbf{r}_\nu-\mathbf{r}_\mu)}, \quad \mathbf{K} = \mathbf{Q}+\mathbf{G}, \tag{78}$$

and the self term is

$$\mathsf{D}^{\text{self}}_{\mu\nu} = -\delta_{\mu\nu}C_{\text{dip}}\frac{4\eta^3}{3\sqrt{\pi}}I_3. \tag{79}$$

The three contributions combine as

$$\mathsf{D}_{\mu\nu}(\mathbf{Q}) = \mathsf{D}^{\text{real}}_{\mu\nu}(\mathbf{Q}) + \mathsf{D}^{\text{recip}}_{\mu\nu}(\mathbf{Q}) + \mathsf{D}^{\text{self}}_{\mu\nu}. \tag{80}$$

The smooth-Γ convention removes the nonanalytic magnetostatic branch with the smallest $|\mathbf{Q}+\mathbf{G}|$. On an extended-zone path this is equivalent to folding into the first Brillouin zone and then dropping the macroscopic $G = 0$ branch. The production parameters are $\eta = 0.15\,\text{Å}^{-1}$, $R_{\text{real}} = 15\,\text{Å}$, and $G_{\text{cut}} = 6\,\text{Å}^{-1}$.

8.3 From the DDI tensor to A_{dip} and B_{dip}

Rotate each sublattice pair,

$$\widetilde{\mathsf{D}}_{\mu\nu}(\mathbf{Q}) = R^\mathsf{T}_\mu \mathsf{D}_{\mu\nu}(\mathbf{Q})R_\nu. \tag{81}$$

The transverse normal and pairing coefficients are

$$t^{\text{dip}}_{\mu\nu}(\mathbf{Q}) = \frac{S}{2}\left[(D_{xx}+D_{yy}) + i(D_{yx}-D_{xy})\right], \tag{82}$$

$$p^{\text{dip}}_{\mu\nu}(\mathbf{Q}) = \frac{S}{2}\left[(D_{xx}-D_{yy}) - i(D_{xy}+D_{yx})\right]. \tag{83}$$

The corresponding BdG blocks are

$$[A_{\rm dip}(\mathbf{Q})]_{\mu\nu} = t^{\rm dip}_{\mu\nu}(\mathbf{Q}) - \delta_{\mu\nu} S \sum_{\rho} [\widetilde{\mathsf{D}}_{\mu\rho}(0)]_{zz}, \tag{84}$$

$$[B_{\rm dip}(\mathbf{Q})]_{\mu\nu} = p^{\rm dip}_{\mu\nu}(\mathbf{Q}). \tag{85}$$

The second term is the longitudinal restoring field about the static collinear state and is evaluated from $\mathsf{D}(0)$. The complete DDI BdG contribution is

$$H^{\rm BdG}_{\rm dip}(\mathbf{Q}) = \begin{pmatrix} A_{\rm dip}(\mathbf{Q}) & B_{\rm dip}(\mathbf{Q}) \\ B^{\dagger}_{\rm dip}(\mathbf{Q}) & A^{*}_{\rm dip}(\mathbf{Q}) \end{pmatrix}. \tag{86}$$

All same-sublattice, inter-sublattice, normal, and anomalous blocks are retained; $D_{\mathbf{Q}}$ is not identified with a single AB tensor element.

9. Complete quadratic Hamiltonian and Colpa diagonalization

Combining all terms gives

$$A(\mathbf{Q}, B) = A_{\rm ex} + A_{\rm SIA} + A_{Z} + A_{\rm dip}, \tag{87}$$

$$B(\mathbf{Q}) = B_{\rm ex} + B_{\rm SIA} + B_{\rm dip}, \tag{88}$$

$$H^{\rm BdG}(\mathbf{Q}, B) = \begin{pmatrix} A(\mathbf{Q}, B) & B(\mathbf{Q}) \\ B^{\dagger}(\mathbf{Q}) & A^{*}(\mathbf{Q}, B) \end{pmatrix}. \tag{89}$$

The quadratic bosonic Hamiltonian is

$$\mathcal{H}^{(2)} = E_{\rm cl} + \frac{1}{2} \sum_{\mathbf{Q}} \Psi^{\dagger}_{\mathbf{Q}} H^{\rm BdG}(\mathbf{Q}) \Psi_{\mathbf{Q}} + \text{const.} \tag{90}$$

The bosonic eigenproblem has the generalized form

$$H^{\rm BdG} u_n = \omega_n \eta u_n, \qquad \eta = \mathrm{diag}(1, 1, -1, -1), \tag{91}$$

or equivalently one diagonalizes $\eta H^{\rm BdG}$. The production calculation uses a Colpa/Cholesky procedure and enforces the paraunitary condition

$$T^{\dagger} \eta T = \eta. \tag{92}$$

The paraunitary error is monitored by

$$\epsilon_{\rm para} = \max |T^{\dagger} \eta T - \eta|. \tag{93}$$

10. Effective two-mode Hamiltonian: projection from 4×4 BdG to 2×2

Near the target momentum, the full 4×4 bosonic BdG problem contains an isolated two-dimensional positive-frequency subspace. Projecting the full Hamiltonian onto this subspace gives the general 2×2 Hermitian effective Hamiltonian

$$H_{\rm eff} = \bar{\omega} I + \frac{1}{2}(\Lambda + h)\sigma_z + d_x \sigma_x + d_y \sigma_y, \tag{94}$$

whose eigenvalue splitting satisfies

$$\Omega^2 = (\Lambda + h)^2 + 4(d_x^2 + d_y^2). \tag{95}$$

The exchange imbalance selects the diabatic basis and fixes the physical meaning of the Pauli components; the exchange, Zeeman, and DDI projections follow.

10.1 Equal-J_7 exchange-only reference doublet

Set

$$J_{7a} = J_{7b} = \bar{J}_7, \tag{96}$$

with DDI, SIA, and field switched off, so that $\lambda_{\rm AM} = 0$. The exchange BdG matrix is then

$$H_0(\mathbf{k}) = \begin{pmatrix} A_0 & 0 & 0 & \Gamma \\ 0 & A_0 & \Gamma & 0 \\ 0 & \Gamma & A_0 & 0 \\ \Gamma & 0 & 0 & A_0 \end{pmatrix}, \qquad \eta = \mathrm{diag}(1, 1, -1, -1). \tag{97}$$

The generalized eigenproblem

$$H_0 u = \omega \eta u \tag{98}$$

decomposes into two identical 2×2 bosonic blocks, $(a_{\mathbf{k}}, b^{\dagger}_{-\mathbf{k}})$ and $(b_{\mathbf{k}}, a^{\dagger}_{-\mathbf{k}})$. The two positive-frequency branches are therefore exactly degenerate,

$$\omega_0(\mathbf{k}) = \sqrt{A_0^2(\mathbf{k}) - \Gamma^2(\mathbf{k})}. \tag{99}$$

A positive-norm basis may be chosen as

$$u_1 = \begin{pmatrix} u \\ 0 \\ 0 \\ v \end{pmatrix}, \qquad u_2 = \begin{pmatrix} 0 \\ u \\ v \\ 0 \end{pmatrix}, \tag{100}$$

with

$$u^2 - v^2 = 1, \qquad v = -\frac{\Gamma}{A_0 + \omega_0} u. \tag{101}$$

Equivalently,

$$u = \sqrt{\frac{1}{2}\left(\frac{A_0}{\omega_0} + 1\right)}, \qquad |v| = \sqrt{\frac{1}{2}\left(\frac{A_0}{\omega_0} - 1\right)}, \tag{102}$$

with the sign of v fixed by Γ. Defining

$$U_0 = (u_1, u_2), \qquad U_0^\dagger \eta U_0 = I_2, \tag{103}$$

makes the target positive-frequency subspace two-dimensional, and the two-mode effective model is the projection of the full BdG problem onto this subspace.

10.2 Projection in the bosonic BdG subspace

Let

$$H = H_0 + \delta H, \qquad \omega = \omega_0 + \delta\omega, \qquad u = U_0 c + \cdots . \tag{104}$$

Substitution into

$$Hu = \omega \eta u \tag{105}$$

and left multiplication by $U_0^\dagger$, using

$$H_0 U_0 = \omega_0 \eta U_0, \qquad U_0^\dagger \eta U_0 = I_2, \tag{106}$$

gives, to first order,

$$U_0^\dagger \delta H U_0 \, c = \delta\omega \, c. \tag{107}$$

The first-order effective operator in the degenerate positive-frequency doublet is

$$M[\delta H] = U_0^\dagger \delta H U_0. \tag{108}$$

The positive-norm normalization already includes the bosonic metric η, so no additional η enters the projected matrix.

10.3 Exchange-imbalance contribution

Restoring the exchange imbalance introduces the normal-block term

$$\delta A_{\mathrm{AM}} = \begin{pmatrix} -\lambda/2 & 0 \\ 0 & +\lambda/2 \end{pmatrix}. \tag{109}$$

In the full Nambu space,

$$\delta H_{\mathrm{AM}} = \mathrm{diag}(-\lambda/2, +\lambda/2, -\lambda/2, +\lambda/2). \tag{110}$$

Projection onto u_1, u_2 gives

$$\langle u_1 | \delta H_{\mathrm{AM}} | u_1 \rangle = -\frac{\lambda}{2} u^2 + \frac{\lambda}{2} v^2 = -\frac{\lambda}{2}(u^2 - v^2) = -\frac{\lambda}{2}, \tag{111}$$

$$\langle u_2 | \delta H_{\mathrm{AM}} | u_2 \rangle = +\frac{\lambda}{2} u^2 - \frac{\lambda}{2} v^2 = +\frac{\lambda}{2}, \tag{112}$$

$$\langle u_1 | \delta H_{\mathrm{AM}} | u_2 \rangle = 0. \tag{113}$$

The projection gives

$$M_{\mathrm{AM}} = -\frac{\lambda}{2}\sigma_z = \frac{\Lambda}{2}\sigma_z, \qquad \Lambda \equiv -\lambda. \tag{114}$$

In the chosen branch ordering, $\Lambda = -\lambda_{\mathrm{AM}}$ is the signed diagonal detuning generated by the exchange imbalance in the positive-frequency doublet.

10.4 Zeeman contribution

The quadratic Zeeman matrix is

$$\delta H_Z = g\mu_B B \ \mathrm{diag}(+1, -1, +1, -1). \tag{115}$$

Projection gives

$$\langle u_1|\delta H_Z|u_1\rangle = +g\mu_B B(u^2 - v^2) = +g\mu_B B, \tag{116}$$

$$\langle u_2|\delta H_Z|u_2\rangle = -g\mu_B B(u^2 - v^2) = -g\mu_B B, \tag{117}$$

$$\langle u_1|\delta H_Z|u_2\rangle = 0. \tag{118}$$

The projection gives

$$M_Z = g\mu_B B\, \sigma_z. \tag{119}$$

The relative diabatic detuning is

$$\Lambda + h(B) = \Lambda + 2g\mu_B B. \tag{120}$$

The Zeeman and exchange detunings act along the same Pauli direction in the same two-dimensional Hilbert space; compensation occurs when their diagonal contributions cancel.

10.5 Projected dipolar mixing

Projecting the complete Ewald DDI BdG operator into the same diabatic basis gives

$$M_{\mathrm{DDI}} = U^\dagger H_{\mathrm{dip}}^{\mathrm{BdG}} U. \tag{121}$$

As a 2×2 Hermitian matrix, M_{DDI} has the unique decomposition

$$M_{\mathrm{DDI}} = d_0 I + d_x\sigma_x + d_y\sigma_y + d_z\sigma_z. \tag{122}$$

Explicitly, if

$$M_{\mathrm{DDI}} = \begin{pmatrix} m_{11} & V \\ V^* & m_{22} \end{pmatrix}, \tag{123}$$

then

$$d_0 = \frac{m_{11} + m_{22}}{2}, \tag{124}$$

$$d_z = \frac{m_{11} - m_{22}}{2}, \tag{125}$$

$$d_x = \Re V, \tag{126}$$

$$d_y = -\Im V. \tag{127}$$

The off-diagonal part defines the transverse mixing,

$$D_{\mathbf{Q}} = 2|V_{\mathbf{Q}}| = 2\sqrt{d_x^2 + d_y^2}. \tag{128}$$

Explicit projection gives $2d_z \simeq 0$ to machine precision, so the DDI contributes only an identity shift and transverse mixing in this basis. A finite d_z can be absorbed into the total diagonal detuning.

10.6 Two-mode eigenvalue relation

Combining the projected terms and absorbing common energy shifts into $\bar{\omega}$ gives

$$H_{\mathrm{eff}} = \bar{\omega} I + \begin{pmatrix} (\Lambda + h + 2d_z)/2 & V \\ V^* & -(\Lambda + h + 2d_z)/2 \end{pmatrix}. \tag{129}$$

For a matrix

$$\begin{pmatrix} z & V \\ V^* & -z \end{pmatrix}, \tag{130}$$

the eigenvalues are

$$\pm\sqrt{z^2 + |V|^2}. \tag{131}$$

The two-level splitting is

$$\Omega = 2\sqrt{\left(\frac{\Lambda + h + 2d_z}{2}\right)^2 + |V|^2}, \tag{132}$$

or

$$\Omega^2 = (\Lambda + h + 2d_z)^2 + D_{\mathbf{Q}}^2. \tag{133}$$

Using the result $2d_z \simeq 0$ above and $h = 2g_{\mathrm{eff}}\mu_B B$ gives

$$\Omega_{\mathbf{Q}}^2(B) = [\Lambda_{\mathbf{Q}} + 2g_{\mathrm{eff}}\mu_B B]^2 + D_{\mathbf{Q}}^2. \tag{134}$$

The sum-of-squares relation follows directly from the eigenvalues of the 2×2 Hermitian effective Hamiltonian. It applies when the target doublet is separated from the remaining positive- and negative-frequency sectors and the Pauli components are identified in the exchange-defined diabatic basis.

11. From the full BdG problem to the target doublet: $\Lambda_{\mathbf{Q}}$ and $D_{\mathbf{Q}}$

11.1 Exchange-defined reference doublet

To distinguish the full-BdG gap from the projected DDI mixing parameter, define the symmetric exchange-only reference

$$J_{7a} = J_{7b} = \bar{J}_7, \tag{135}$$

with DDI and field switched off while all other exchange couplings are kept unchanged:

$$H_0(\mathbf{Q}) = H_{\mathrm{ex}}^{\mathrm{BdG}}(\mathbf{Q}; J_{7a} = J_{7b} = \bar{J}_7). \tag{136}$$

At the target momentum, the two positive-frequency reference modes are numerically degenerate. Let

$$U_0 = (u_1, u_2), \qquad U_0^\dagger \eta U_0 = I_2. \tag{137}$$

For a small BdG perturbation δH, the projected matrix in the degenerate positive-frequency subspace is

$$M[\delta H] = U_0^\dagger \delta H \, U_0. \tag{138}$$

This uses the positive-norm projection rule established above.

11.2 Fixing the diabatic basis with the exchange imbalance

Define

$$\delta H_{\mathrm{AM}}(\mathbf{Q}) = H_{\mathrm{ex}}^{\mathrm{split}}(\mathbf{Q}) - H_0(\mathbf{Q}). \tag{139}$$

The exchange-imbalance projection is

$$M_{\mathrm{AM}} = U_0^\dagger \delta H_{\mathrm{AM}} U_0. \tag{140}$$

Because H_0 is degenerate, arbitrary linear combinations of u_1, u_2 are allowed. Diagonalizing the 2×2 Hermitian matrix M_{AM} fixes the diabatic basis,

$$U = U_0 W, \tag{141}$$

with the two diabatic branches ordered according to the convention

$$\Lambda_{\mathbf{Q}} = -\lambda_{\mathrm{AM}}(\mathbf{Q}). \tag{142}$$

Then

$$U^\dagger \delta H_{\mathrm{AM}} U = \begin{pmatrix} \Lambda_{\mathbf{Q}}/2 & 0 \\ 0 & -\Lambda_{\mathbf{Q}}/2 \end{pmatrix} \tag{143}$$

up to an identity shift that does not affect the splitting.

11.3 Explicit projection of the complete DDI

In the same exchange-defined basis, the Ewald DDI operator projects to

$$M_{\mathrm{DDI}}(\mathbf{Q}) = U^\dagger H_{\mathrm{dip}}^{\mathrm{BdG}}(\mathbf{Q}) U. \tag{144}$$

Using the Pauli decomposition above,

$$M_{\mathrm{DDI}} = d_0 I + d_x \sigma_x + d_y \sigma_y + d_z \sigma_z. \tag{145}$$

If

$$V_{\mathbf{Q}} = [M_{\mathrm{DDI}}]_{12} = d_x - i d_y, \tag{146}$$

then

$$D_{\mathbf{Q}} = 2|V_{\mathbf{Q}}| = 2\sqrt{d_x^2 + d_y^2}. \tag{147}$$

The projected diagonal DDI detuning is

$$\Lambda_{\mathrm{DDI}}^{\mathrm{diag}} = 2 d_z \tag{148}$$

and is evaluated explicitly at the momentum of interest.

The individual d_x, d_y depend on the phases chosen for the two diabatic basis vectors. Under

$$u_1 \to e^{i\phi_1} u_1, \qquad u_2 \to e^{i\phi_2} u_2, \tag{149}$$

$V_{\mathbf{Q}}$ acquires a phase, whereas

$$D_{\mathbf{Q}} = 2|V_{\mathbf{Q}}| \tag{150}$$

remains invariant. Hence $D_{\mathbf{Q}}$ is the phase-gauge-invariant mixing scale, whereas d_x and d_y separately have no independent physical meaning.

11.4 Numerical validation of the projected parameter

At the working momentum $\mathbf{Q}_A = (0.25, 0.75, 0)$, explicit projection gives

$$D_A^{\text{proj}} = 121.809730\,\mu\text{eV}, \tag{151}$$

whereas the equal-J_7 full-BdG avoided-crossing floor including the complete DDI is

$$\Omega_{\text{eq}}^{\text{full}}(0) = 121.592033\,\mu\text{eV}. \tag{152}$$

The difference is $0.217697\,\mu$eV, or 0.179%, which measures higher-order corrections from Nambu sectors outside the target doublet; the projected parameter and the complete full-BdG floor differ at the sub-percent level.

12. Two-mode formula and compensation field

Neglecting the projected diagonal DDI correction, the target doublet is described by

$$H_{\text{eff}}(\mathbf{Q}, B) = \bar{\omega}_{\mathbf{Q}} I + \frac{1}{2}[\Lambda_{\mathbf{Q}} + 2g_{\text{eff}}\mu_{\text{B}}B]\sigma_z + d_x\sigma_x + d_y\sigma_y. \tag{153}$$

The eigenvalue splitting satisfies

$$\boxed{\Omega_{\mathbf{Q}}^2(B) = [\Lambda_{\mathbf{Q}} + 2g_{\text{eff}}\mu_{\text{B}}B]^2 + D_{\mathbf{Q}}^2}. \tag{154}$$

If a diagonal DDI correction is retained, replace $\Lambda_{\mathbf{Q}}$ by

$$\Lambda_{\mathbf{Q}}^{\text{tot}} = \Lambda_{\mathbf{Q}} + 2d_z. \tag{155}$$

The minimum splitting occurs at

$$\boxed{B^*(\mathbf{Q}) = -\frac{\Lambda_{\mathbf{Q}}}{2g_{\text{eff}}\mu_{\text{B}}}}. \tag{156}$$

Here $D_{\mathbf{Q}}$ fixes the minimum avoided-crossing gap, while the symmetry-odd exchange detuning fixes its position along the field axis.

At zero field,

$$\Omega_{\mathbf{Q}}(0) = \sqrt{\Lambda_{\mathbf{Q}}^2 + D_{\mathbf{Q}}^2}. \tag{157}$$

For $|\Lambda| \ll D$,

$$\Omega(0) - D \simeq \frac{\Lambda^2}{2D}, \tag{158}$$

The weak exchange enters the zero-field splitting in an even, quadratic form, whereas B^* is linear in Λ and signed.

13. Symmetry-related momentum partners and fixed-field contrast

Take

$$\mathbf{Q}_A = (0.25, 0.75, 0), \qquad \mathbf{Q}_{A'} = (-0.75, 0.25, 0) = C_{4z}\mathbf{Q}_A. \tag{159}$$

From

$$\lambda_{\text{AM}}(\mathbf{Q}_A) = -4S\delta J_7, \qquad \lambda_{\text{AM}}(\mathbf{Q}_{A'}) = +4S\delta J_7, \tag{160}$$

we obtain

$$\Lambda_A = +4S\delta J_7, \qquad \Lambda_{A'} = -4S\delta J_7. \tag{161}$$

The full-BdG symmetry gives

$$\Omega_{A'}(B) = \Omega_A(-B), \tag{162}$$

and hence

$$B_A^* = -B_{A'}^* = -\frac{2S\delta J_7}{g_{\text{eff}}\mu_{\text{B}}}. \tag{163}$$

Define

$$C(B) = \Omega_A^2(B) - \Omega_{A'}^2(B). \tag{164}$$

If the projected partner mixing satisfies $D_A^2 = D_{A'}^2$, the common mixing term cancels, giving

$$C(B) = 8g_{\text{eff}}\mu_{\text{B}}\Lambda_A B, \tag{165}$$

with slope

$$\frac{dC}{dB} = 8g_{\text{eff}}\mu_{\text{B}}\Lambda_A = 32Sg_{\text{eff}}\mu_{\text{B}}\delta J_7. \tag{166}$$

14. Conversion of D_c in the FeF_2 transfer calculation

The FeF_2 literature model uses the axial single-ion term

$$\mathcal{H}_{\rm SIA}^{\rm lit} = -D \sum_i (S_i^z)^2, \qquad D = 0.82\,\text{meV}, \qquad S = 2. \tag{167}$$

The code uses the convention

$$\mathcal{H}_{\rm SIA}^{\rm code} = D_c \sum_i (S_i^z)^2, \tag{168}$$

with harmonic HP contribution

$$A_{\rm SIA}^{\rm axial} = -2SD_c. \tag{169}$$

The Hamiltonian sign convention gives $D_c = -D$. To reproduce the spin-wave energy scale associated with the published FeF_2 value $D = 0.82$ meV in the LSWT/BdG convention, we use the finite-S conversion

$$D_c^{\rm core} = -D\left(1 - \frac{1}{2S}\right). \tag{170}$$

For $S = 2$,

$$D_c^{\rm core} = -0.82\left(1 - \frac{1}{4}\right) = -0.615\,\text{meV}. \tag{171}$$

The factor follows from the finite-S level spacing of a single spin. For $-D(S^z)^2$, the exact excitation energy from $m = S$ to $m = S - 1$ is

$$\Delta E_{S\to S-1} = D\left[S^2 - (S-1)^2\right] = D(2S-1). \tag{172}$$

The harmonic HP coefficient has scale $2S|D_c^{\rm core}|$. Equating the two gives

$$2S|D_c^{\rm core}| = D(2S-1), \tag{173}$$

and hence

$$|D_c^{\rm core}| = D\left(1 - \frac{1}{2S}\right). \tag{174}$$

The factor $1 - 1/(2S)$ maps the finite-S single-ion level spacing onto the harmonic bosonic coefficient and is used only as a convention conversion, without introducing an additional interaction or fitted parameter.